Business Requirements Document (BRD)

Project: Customer & Order Data Ingestion, SCD Type 2 Orders Summary, and Customer Aggregate Spend

Prepared by: Senior Business Analyst

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1. Executive Summary

- Purpose: Define business and functional requirements to read customer and order CSV data, cleanse and transform it, persist as Delta tables, create an SCD Type 2 orders summary, and produce daily customer aggregate spend. The solution will be implemented using Databricks Delta Lake and Delta Live Tables (DLT).
- Goals:
- Reliable ingestion of customer and order CSV sources into Delta format.
- Standardized schemas for customer, order, ordersummary, and customeraggregatespend tables.
- Data quality rules: remove null and "Null" values and deduplicate.
- Calculate TotalAmount for orders.
- Maintain ordersummary as SCD Type 2 to preserve customer history.
- Produce daily aggregated spend per customer (Name, Date, TotalAmount).
- Automate the pipeline using Delta Live Tables.

2. Stakeholders

- Business Owner: Sales/Revenue Analytics team
- Data Owner: Data Engineering team (Databricks / Lakehouse)
- Consumers: BI/Reporting, Finance, Marketing, Customer Success
- Compliance & Security: Data Governance, Security teams
- Implementation Team: Data Engineers, ETL Developers, QA Engineers

3. Scope

3.1 Included

- Read CSV files from specified volumes/paths for customer and order datasets.
- Create/overwrite Delta tables for customer and order.
- Apply cleansing rules (null / "Null" removal, deduplication).
- Add TotalAmount to order records.
- Create ordersummary table in catalog gen_ai_poc_databrickscoe.sdlc_wizard if not exists.
- Join customer and order on CustId and load into ordersummary with SCD Type 2 semantics.

- Implement logic to update SCD Type 2 ordersummary when customer attributes change.
- Create customeraggregatespend table and populate it from aggregated ordersummary.
- Implement all processing using Delta Live Tables for automated orchestration, lineage, and monitoring.

3.2 Excluded

- Ingestion from sources other than the provided CSV file paths.
- Upstream changes to CSV schema outside specified fields (unless change control raises a requirement).
- Real-time streaming ingestion (current requirement is file-based batch via DLT).
- Downstream visualization and dashboard creation (outside scope).

4. Assumptions

- The CSV files for customer and order reside at:
- /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata
- /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata
- Files are readable by Databricks workspace and appropriate IAM and storage mounts are configured.
- The provided schemas are the authoritative fields for ingestion. Any additional columns will be handled via change control.
- Timestamps/timezone handling for Date fields will be standardized (see Non■functional).
- Name is a sufficiently stable identifier for aggregation (customer name may not be unique; business to confirm if aggregation should use CustId instead).
- Delta Live Tables runtime and required cluster resources are available.

5. Requirements

5.1 Functional Requirements (FR)

FR-1: Ingest source CSV data

- FR-1.1: Read customer CSV files from /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata and write to Delta table "customer".
- FR-1.2: Read order CSV files from /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata and write to Delta table "order".

FR-2: Source schemas

- FR-2.1: customer schema must include fields: CustId, Name, EmailId, Region.
- FR-2.2: order schema must include fields: OrderId, ItemName, PricePerUnit, Qty, Date, CustId.

FR-3: Data enrichment

- FR-3.1: Compute TotalAmount on order as PricePerUnit * Qty and add as new column TotalAmount.

FR-4: Data quality cleansing

- FR-4.1: Remove records where any required field is null or the literal string "Null".
- FR-4.2: Remove duplicate records in both customer and order tables. Deduplication strategies must be defined per table (see Design & Rules).

FR-5: Orders summary table

- FR-5.1: Create table ordersummary in catalog gen_ai_poc_databrickscoe, schema sdlc_wizard if it does not exist.
- FR-5.2: ordersummary must contain fields: CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date, TotalAmount (TotalAmount is implied from step 3).

FR-6: SCD Type 2 behavior for ordersummary

- FR-6.1: ordersummary must be maintained as SCD Type 2.
- FR-6.2: When a customer record changes (e.g., Name, EmailId, Region), existing records related to that customer should be marked Inactive and assigned an EndDate.
- FR-6.3: Insert new records for the new customer state as Active with StartDate.
- FR-6.4: Maintain history for all changes; do not delete historical rows.

FR-7: Customer aggregate spend

- FR-7.1: Create table customeraggregatespend in catalog gen_ai_poc_databrickscoe, schema sdlc_wizard if not exists, with columns: Name, TotalAmount, Date.
- FR-7.2: Populate customeraggregatespend by aggregating ordersummary: GROUP BY Name, Date with SUM(TotalAmount) as TotalAmount.

FR-8: Implementation orchestration

- FR-8.1: Implement steps FR-1 through FR-7 using Delta Live Tables (DLT) pipelines.
- FR-8.2: Pipeline should support incremental processing and be idempotent.

5.2 Non■Functional Requirements (NFR)

NFR-1: Performance

- NFR-1.1: Pipeline must complete daily batch within a defined SLA (to be agreed — e.g., within 1 hour of data availability).
- NFR-1.2: Aggregations should be optimized using partition pruning (e.g., partition by Date) and suitable file sizes per Delta best practices.

NFR-2: Reliability and Monitoring

- NFR-2.1: DLT must provide visibility, metrics, and alerts on pipeline failures, data quality rule violations, and throughput.
- NFR-2.2: Implementation must support retries and resumability.

NFR-3: Data Retention & History

- NFR-3.1: SCD Type 2 ordersummary must preserve full history with StartDate and EndDate fields for each record; default Active flag indicates current records.
- NFR-3.2: Retention policy for historical records should be determined by data governance (e.g., keep 7 years) and implemented as a separate archival process.

NFR-4: Security & Access

- NFR-4.1: Only authorized roles can read/write the target catalog gen_ai_poc_databrickscoe.sdlc_wizard.
- NFR-4.2: Sensitive fields (e.g., EmailId) must follow data privacy rules; encryption at rest and in transit required.
- NFR-4.3: Audit logging for changes to SCD table required.

NFR-5: Data Format & Type Standards

- NFR-5.1: Standardize Date to ISO-8601 (YYYY-MM-DD) or timestamp with timezone; specify timezone handling.
- NFR-5.2: PricePerUnit, Qty, TotalAmount should be stored as appropriate numeric types (e.g., DECIMAL(18,4) for currency).
- NFR-5.3: CustId and OrderId types defined consistently (string or numeric) — choose string to avoid leading-zero issues unless business confirms numeric.

6. Detailed Design & Business Rules

6.1 Source ingestion

- Use DLT to create two streaming or batch tables (depending on file arrival): Bronze stage Delta tables "customer_bronze" and "order_bronze" from CSV source paths. Include ingestion metadata columns (e.g., _ingest_time, _file_name) for traceability.

6.2 Schema enforcement & parsing

- Enforce expected column list; if extra columns exist, log and drop them unless change request is approved.
- Implement parsing rules: trim strings, normalize casing for Region, and validate numeric and date formats.

6.3 Data quality rules

- Null handling:
- Define required fields:

- customer: CustId (required), Name (required), EmailId (optional/required per business), Region (optional).
- order: OrderId (required), PricePerUnit (required), Qty (required), Date (required), CustId (required).
- Remove rows where any required field is null, empty string, or the literal "Null" (case-insensitive).
- Log rejected rows to a quarantine/error Delta table with reason codes.
- Deduplication:
- customer dedupe key: CustId (primary); if multiple records for same CustId in same batch, choose most recent by ingest timestamp or last-modified column; log duplicates.
- order dedupe key: OrderId; if duplicates, keep single authoritative record (e.g., latest ingest), log duplicates.

6.4 Calculations

- TotalAmount = PricePerUnit * Qty; compute with numeric precision and round per currency rules.
- Ensure null-safe multiplication (treat null PricePerUnit or Qty as reject per data quality).

6.5 Ordersummary SCD Type 2 design

- Ordersummary table schema (suggested final physical schema):
- CustId (PK candidate)
- Name
- EmailId
- Region
- OrderId
- ItemName
- PricePerUnit
- Qty
- Date (order date)
- TotalAmount
- StartDate (timestamp) — when this record became active
- EndDate (timestamp) — when this record ceased to be active; NULL for active
- IsActive (boolean) — current record indicator
- RecordHash or ChangeHash (optional) — to detect changes efficiently
- SourceMetadata (optional) — e.g., _file_name, _ingest_time

- SCD Type 2 logic:
- When new orders arrive joined with current customer state:
- If the combination of business key (CustId + OrderId) does not exist, insert a new record with StartDate = processing date/time, EndDate = NULL, IsActive = true.

- If the combination exists but customer attributes (Name, EmailId, Region) differ from the current active record (detected via RecordHash or field comparison), then:
- Update existing active record: set EndDate = processing timestamp, IsActive = false.
- Insert new record with updated customer attributes and StartDate = processing timestamp, EndDate = NULL, IsActive = true.
- If only order-level details changed (Price, Qty, ItemName), evaluate whether to treat as new version; by default, preserve history for customer attribute changes primarily. Business to confirm whether order edits create new SCD rows.
- Implement SCD updates using Delta MERGE or DLT-managed expectation/streams to ensure idempotency.

6.6 Aggregation to customeraggregatespend

- Aggregation logic:
- Source: ordersummary effective rows (including only active or all rows as per business rule — default use Date field in ordersummary).
- Group by Name and Date (Date should be truncated to date granularity).
- SUM(TotalAmount) as TotalAmount.
- Target table customeraggregatespend schema:
- Name
- Date
- TotalAmount
- Partition and indexing:
- Partition by Date for performance.
- If Name cardinality high, consider clustering instead of partitioning by Name.

6.7 Delta Live Tables implementation

- Build DLT pipeline with following tables/views:
- customer_bronze (raw ingest)
- order_bronze (raw ingest)
- customer_clean (quality-checked, deduped)
- order_clean (quality-checked, deduped, TotalAmount added)
- ordersummary_scd (SCD Type 2 physical table)
- customeraggregatespend (materialized aggregation)
- Use DLT expectations for data quality rules (null checks, uniqueness) and route violations to quarantine.
- Leverage DLT incremental model: use streaming/file-based auto-ingest to handle new files.
- Use DLT's plan or merge transformations to manage SCD Type 2 merges.

7. Data Validation & Test Cases

- Ingestion tests:
- Provide sample CSVs with normal and anomalous rows. Confirm that valid rows are loaded to delta and invalid rows to quarantine.
- TotalAmount tests:
- Cases with integer and decimal PricePerUnit and Qty; validate computed TotalAmount and rounding.
- Null & "Null" tests:
- Rows containing literal "Null" in different cases and whitespace; confirm they are rejected if in required fields.
- Deduplication tests:
- Duplicate customer rows in same batch; verify retention of single record and logging.
- Duplicate order rows; verify OrderId uniqueness enforcement.
- SCD Type 2 tests:
- Update a customer's Name or Region and ingest; confirm previous row EndDate is set and new active row created.
- Insert order for existing and new customers; validate SCD behavior.
- Aggregation tests:
- Multiple orders per Name and Date; confirm SUM(TotalAmount) accuracy.

8. Error Handling & Logging

- Quarantine tables for customer_rejected and order_rejected with fields: raw_row, reason_code, reason_message, ingest_metadata.
- Operational logging for DLT: pipeline start/stop, failures, table-level metrics (rows read, rows written, expectations failed).
- Alerts: Notify Data Engineering and Business Owners on pipeline failures and high rejection rates.

9. Security, Compliance, and Governance

- Access control using Unity Catalog (or equivalent) to restrict operations on gen_ai_poc_databrickscoe.sdlc_wizard.*.
- Encryption at rest and in transit as per organizational policy.
- PII handling: Flag EmailId as sensitive; mask or limit access in downstream datasets unless authorized.
- Data lineage: DLT provides lineage; ensure it’s captured for audits.

10. Deployment & Operational Considerations

- Environments: dev, test, staging, prod with separate catalogs/schemas or prefixes.
- CI/CD: Store DLT notebook or pipeline definitions in Git; CI/CD to deploy pipeline changes across environments.

- Schedule: DLT pipeline trigger schedule (e.g., every hour or daily); configurable.
- Resource sizing: Determine cluster sizes and autoscaling policies to meet SLA.
- Backups & Recovery: Delta table backups/versioning and time travel retention policies configured.

11. Open Decisions & Questions

- Q1: Should customeraggregatespend group by CustId instead of Name to avoid ambiguity with duplicate names? Recommendation: Use CustId + Name and preserve display Name for reports; business to confirm.
- Q2: Which fields constitute a customer change that triggers SCD Type 2? (Name, EmailId, Region — confirm whether other attributes may be added).
- Q3: Should order edits (e.g., PricePerUnit or Qty change) create new SCD rows for ordersummary, or should order-level changes be handled separately? Business to confirm.
- Q4: Date timezone: which timezone to use for Date field normalization? (UTC recommended).
- Q5: Retention policy and archival timeline for historical SCD rows.
- Q6: Are EmailId and other PII permitted in aggregated table customeraggregatespend? If not, consider hashing or excluding.

12. Data Model Summary (logical)

- customer (Delta)
- CustId (PK)
- Name
- EmailId
- Region
- _ingest_time, _file_name (metadata)
- order (Delta)
- OrderId (PK)
- ItemName
- PricePerUnit (DECIMAL)
- Qty (INT/DECIMAL as needed)
- Date (DATE or TIMESTAMP)
- CustId (FK to customer)
- TotalAmount (DECIMAL)
- _ingest_time, _file_name (metadata)
- ordersummary (SCD Type 2 Delta)
- CustId
- Name
- EmailId

- Region
- OrderId
- ItemName
- PricePerUnit
- Qty
- Date
- TotalAmount
- StartDate
- EndDate
- IsActive
- RecordHash
- SourceMetadata
- customeraggregatespend (Delta)
- Name
- Date
- TotalAmount

13. Implementation Roadmap (high-level)

- Week 0–1: Requirements sign-off, access approvals, environment setup.
- Week 1–2: DLT pipeline design and small POC ingest of CSV -> bronze -> clean.
- Week 2–3: Implement cleansing, TotalAmount calculation, dedupe, and tests.
- Week 3–4: Implement SCD Type 2 logic (ordersummary) and test cases.
- Week 4–5: Implement aggregation to customeraggregatespend and performance tuning.
- Week 5–6: User acceptance testing, security review, documentation, and production deployment.

14. Acceptance Criteria

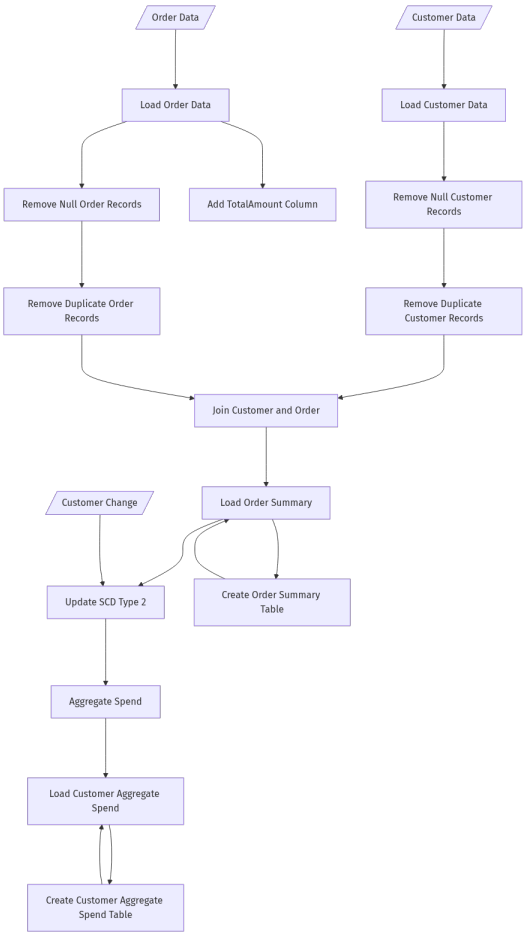
- Source CSV files are ingested into Delta customer and order tables at configured cadence.
- TotalAmount computed correctly for all order records.
- No records with null/ "Null" in required fields exist in production customer and order tables; rejected records visible in quarantine.
- Duplicate records removed per business rules; dedupe logs are available.
- ordersummary exists as SCD Type 2 table; when customer attributes change, historical records have EndDate and IsActive = false, and new rows are inserted with StartDate and IsActive = true.
- customeraggregatespend contains correct daily aggregated TotalAmount by Name and Date.
- All processing implemented with Delta Live Tables, observable in DLT UI with expectations and metrics.

15. Appendix

- Source file paths:
- customer: /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata
- order: /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata
- Target catalog/schema:
- gen_ai_poc_databrickscoe.sdlc_wizard
- Provided source schemas:
- customer: CustId, Name, EmailId, Region
- order: OrderId, ItemName, PricePerUnit, Qty, Date, CustId
- Aggregation:
- GROUP BY Name, Date
- SUM(TotalAmount) AS TotalAmount
- Implementation technology:
- Delta Lake, Delta Live Tables, Databricks Delta MERGE

If you approve this BRD, next steps are: confirm open decisions (Section 11), finalize retention and SLA values, and begin environment provisioning and DLT pipeline development.

Data Flow Diagram (DFD)


Order Data
Customer Data
Load Order Data
Load Customer Data
Remove Null Order Records
Add TotalAmount Column
Remove Null Customer Records
Remove Duplicate Order Records
Remove Duplicate Customer Records
Join Customer and Order
Customer Change
Load Order Summary
Update SCD Type 2
Create Order Summary Table
Aggregate Spend
Load Customer Aggregate Spend
Create Customer Aggregate Spend Table